

MATTIA RUMOLO

Milan, Italy | mattia.rumolo@mail.polimi.it
[linkedin.com/in/mattia-rumolo](https://www.linkedin.com/in/mattia-rumolo)

PROFESSIONAL SUMMARY

Dedicated Engineering Physics Master's candidate at Politecnico di Milano, specializing in **Micro and Nano-systems** (track co-developed with **STMicroelectronics**). Proven expertise in end-to-end **nano-fabrication** workflows within ISO-classified cleanrooms, specializing in 2D material device physics, **semiconductor physics** and **microelectronics**. Currently developing a smart wearable device on ultra-low-power ARM architectures for intelligent sensory systems.

EDUCATION

- M.Sc. in Engineering Physics – Micro and Nano-systems Track** Sept. 2025 – April 2027 (Exp.)
Politecnico di Milano *Milan, Italy*
- Enrolled in the curriculum co-developed with **STMicroelectronics**, focusing on extreme miniaturization, physical modeling of solid-state devices, nano-fabrication and advanced MEMS integration.
 - **Core Coursework:** Condensed Matter Physics, Semiconductor Device Physics and Microelectronics, Physical Modelling and Simulation, Advanced nano-fabrication Technologies, advanced project work.
- B.Sc. in Physics Engineering** Sept. 2022 – Sept. 2025
Politecnico di Milano *Milan, Italy*
- Focus on quantum mechanics, optics, and material science. Developed logical-analytical skills through experimental laboratory practice.
 - **Thesis:** Fabrication and characterization of integrated plasmonic structures on an MoS₂ transistor.

TECHNICAL SKILLS

Nano-fabrication & Cleanroom: ISO 6/8 Protocols, Electron Beam Lithography (EBL - Raith Voyager), Reactive Ion Etching (RIE), Physical Vapor Deposition (PVD), Atomic Layer Deposition (ALD), Spin Coating, Chemical Safety.

Device Physics & Materials: Solid-State Physics, Advanced Quantum Mechanics, 2D Materials (MoS₂, hBN, Graphene), FET Architecture, MEMS Design.

Electronics: Operational amplifier circuit design, Signal recovery and Noise filtering, Noise Analysis.

Embedded Systems: STM32U5 Architecture (Cortex-M33), STM32 firmware programming, Low-Power Background Autonomous Mode (LPBAM), PCB design, Arduino and Arduino IDE, 3D printing, Fast Prototyping.

Software & Data Science: C, Embedded C, Cube MX, VS Code, Altium Designer, Python, MATLAB, Git, LaTeX, Microsoft 365 suite.

ACADEMIC RESEARCH & ENGINEERING PROJECTS

- B.Sc. Thesis: Plasmon-Enhanced 2D MoS₂ Transistor R&D | L-NESS & PoliFab** May 2024 – Sept. 2024
- Engineered a 2D heterostructure FET (**graphene** back-gate, **hBN**, **MoS₂** channel) with Au plasmonic structures to improve responsivity to light. Conducted **EBL**, **RIE**, and EBA in ISO 6 cleanroom.
- Smart Sensory Eyewear | EssilorLuxottica Smart Eyewear Lab** Feb. 2025 – Present
- Currently developing STM32U5 firmware interfacing a AS7341 spectrometer and digital MEMS mic with an **STM32U5** MCU for a prototype of smart glasses; optimizing power via **LPBAM** for Stop 2 mode acquisition.
 - Designing Embedded C algorithms for real-time environmental hazard detection.
- EchoFlow: Ultrasonic Traffic Monitoring System** Sept. 2024 – Jan. 2025
- Designed a distributed traffic-sensing network using **Arduino** and ultrasonic sensors. Employed noise filtering techniques (scalar Kalman filter) and achieved velocity estimation error < 20%.

LEADERSHIP & VOLUNTEERING

- Student Mentor - Buddy Project** Aug. 2024 – Feb. 2025
Politecnico di Milano *Milan, Italy*
- Mentored incoming international students, facilitating their academic and social integration into the university community while managing logistics and administrative orientation.

LANGUAGES

Italian: Native | **English:** C1 (Cambridge certified, Full Professional Proficiency)